

A Latent-Guided Hybrid Architecture for Liver Segmentation in Contrast-Enhanced MRI

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Abstract. Liver segmentation in hepatobiliary-phase (GED4) MRI is crucial for fibrosis assessment but remains challenging due to the scarcity of annotated data. Recent advances in large language models (LLMs), such as LLaMA, have demonstrated remarkable generalization capabilities in learning transferable latent representations across domains. In this work, we introduce LLaMba-Seg, a framework that integrates frozen LLaMA into a selective state-space U-Mamba, forming a hybrid architecture for robust liver segmentation. Our model consists of a trainable U-Mamba encoder-decoder, with a frozen LLaMA module at the bottleneck, which distills high-level semantic context and guides representation alignment. U-Mamba captures spatial details at multiple resolutions for precise boundary delineation in low-contrast settings, while LLaMA’s contextual embeddings enable the creation of high-confidence pseudo-labels from unannotated GED4 scans. Focusing on the contrast-enhanced subtask of segmenting the GED4 sequence, we train the model on a combined dataset that includes the challenge’s GED4 data along with external AMOS and ATLAS data to enhance domain robustness. On the Liver Fibrosis Quantification and Analysis Task 2 (LiSeg) validation benchmark, our model achieves a Dice score of 0.97 and a Hausdorff Distance of 21.34 mm, demonstrating its effectiveness under limited supervision.

Keywords: Liver Segmentation · Hepatobiliary-Phase MRI · Contrast-Enhanced MRI · Latent Representation Transfer · State-Space Models · U-Mamba · Domain Robustness · Limited Supervision.

1 Introduction

The accurate segmentation of the liver from hepatobiliary-phase magnetic resonance imaging (MRI) is a critical prerequisite for assessing liver function and diagnosing hepatic fibrosis [24]. This task is hindered by a confluence of challenges: low-contrast boundaries, high anatomical variability, scarcity of annotated data [2, 13, 23], and the inherent difficulty of learning discriminative latent representations from such limited and complex information [4, 17].

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To address these challenges, previous research [20, 31] has primarily focused on refining deep learning architectures and data augmentation strategies. Aggressive augmentation methods such as elastic deformations and contrast adjustments have been employed to artificially expand limited datasets and improve robustness to anatomical variability. Many other approaches have similarly relied on the U-Net/U-Mamba [20, 25] architecture, leveraging its powerful encoder-decoder structure to capture both contextual information and precise spatial details [25]. These methods still struggled to learn fine-grained latent space representations, often capturing only coarse, high-level features of the anatomy.

Recent advancements in large language models (LLMs), such as LLaMA [32], have showcased remarkable capabilities in learning transferable, high-level latent representations that generalize effectively across various domains [26]. Inspired by this, we propose a novel hybrid architecture that synergistically combines the strengths of vision-specific models with the powerful semantic understanding of LLMs for medical image segmentation.

In this work, we introduce LLaMba-Seg (LLaMA-enhanced Mamba for Segmentation), a framework that integrates a frozen LLaMA model [5, 32] into a selective state-space U-Mamba [20] architecture. Our model leverages a trainable U-Mamba encoder-decoder to capture fine-grained spatial details at multiple resolutions, enabling precise boundary delineation in challenging low-contrast settings. A frozen LLaMA layer is strategically placed between the encoder and decoder to distill high-level semantic features from the image embeddings. This allows the model to leverage LLaMA’s rich contextual understanding to guide representation alignment and, crucially, to generate high-confidence pseudo-labels from unannotated scans.

We validated our approach on the Liver Fibrosis Quantification and Analysis (LiQA) Task 2 Liver Segmentation (LiSeg) challenge’s contrast-enhanced sub-task of segmenting the GED4 sequence. To improve the model’s domain robustness, we employed a multi-source training strategy, utilizing a composite dataset that combines the challenge’s dataset [6, 18, 29] with external scans from the AMOS [13] and ATLAS [23] datasets. Strong performance is obtained on the LiQA Task 2 validation benchmark, where our model achieved a Dice score of 0.97 and a Hausdorff Distance of 21.34 mm. These results underscore the effectiveness of our hybrid architecture under limited supervision and cross-domain variability.

2 Related Work

Backbone of medical image segmentation. U-Net popularized the symmetric encoder-decoder with skip connections for precise localization under limited supervision [25]. Building on this idea, nnU-Net standardized preprocessing, architecture, training and postprocessing into a self-configuring pipeline that generalizes across datasets [12]. To overcome the locality of convolutions and better capture long-range dependencies, hybrid and pure-Transformer designs were introduced: TransUNet couples a CNN with a ViT encoder [3], and UNETR

adopts a pure Transformer encoder for 3D volumes with U-Net style decoding [9]. The quadratic complexity of self-attention, however, is costly for high-resolution 3D inputs. State-space models (SSMs) address this with linear-time sequence modeling. Mamba provides selective SSM blocks with high throughput and linear scaling [8]. Its visual adaptation, VMamba, implements visual state-space blocks that efficiently aggregate global context [19]. In medical imaging, U-Mamba integrates convolutional and Mamba blocks within a U-shaped hierarchy, reporting strong accuracy with favorable memory/speed trade-offs [20]. Concurrent 3D variants such as SegMamba further validate SSMs for volumetric segmentation [30].

LLM/LLaMA layers for medical vision. A complementary line of work leverages large language models (LLMs) as frozen global priors via lightweight adapters. BLIP-2 bridges frozen image encoders and frozen LLMs using a small querying transformer, reducing trainable parameters for vision-language pre-training [16]. LLaMA-Adapter and its V2 variant efficiently inject instructional or visual tokens into a frozen LLaMA through prompts/adapters and early-fusion strategies [5, 32]. In biomedicine, LLaVA-Med adapts a conversational VLM to biomedical imagery for open-ended VQA [15], and Med-Flamingo enables few-shot medical VQA via continued multimodal pretraining [22]. Beyond VQA, a frozen LLM transformer block can be inserted directly as an encoder layer for purely visual biomedical tasks, acting as a plug-and-play global regularizer without prompts [14]. Our method follows this parameter-efficient philosophy: we place a frozen LLaMA adapter at the bottleneck of a U-Mamba and use its output as the deepest skip connection, injecting a globally consistent prior at the coarsest scale while preserving linear-time long-range modeling in the encoder.

3 Method

3.1 Data and preprocessing

We train on Liver Fibrosis Quantification and Analysis (LiQA) dataset [6, 18, 29] hepatobiliary-phase (GED4) annotations (30 cases) and incorporate two external MRI sources for supervision: the ATLAS [23] training subset (60 MRI cases; liver label retained) and the AMOS [13] Task-2 MRI training split (60 MRI cases; liver label extracted from the multi-organ masks). All images and labels are harmonized with SimpleITK by resampling to 1.5 mm isotropic spacing (BSpline for images, nearest-neighbor for labels) while preserving the original origin and direction. We subsequently standardize intensities per volume and apply foreground cropping to reduce empty background.

3.2 Network architecture: U-Net + Mamba + LLaMA

Our model, named LLaMba-Seg, is a 3D U-Mamba encoder-decoder augmented with a frozen LLaMA-2 (7B) [27] adapter at the bottleneck (Fig.1). Each encoder stage stacks two residual blocks [10] (Convolution + InstanceNorm [28]

+ LeakyReLU [21]) followed by a Mamba block [8], which provides linear-time long-range dependency modeling. In the Mamba block, the 3D feature map is first flattened and layer-normalized [1], then processed along two parallel paths: (i) a sequence path that applies a linear projection, depthwise 1D convolution, SiLU [11], and a selective state-space (SSM) update [7]; and (ii) a gating path with a linear projection and SiLU. The two outputs are fused via an element-wise (Hadamard) product, projected back to the original feature width, reshaped to 3D, and added through a residual skip connection. Compared with self-attention, this design aggregates global context with linear time and memory complexity, making it well suited for high-resolution volumes.

At the bottleneck, we insert a frozen LLaMA adapter. The encoder’s deepest features are linearly projected to LLaMA’s hidden width, passed through the frozen transformer stack, and projected back to the encoder width. The resulting LLaMA-conditioned feature serves as deepest connection into the decoder, which then concatenates it with the upsampled decoder features as in U-Net. This design injects globally regularized context while keeping LLaMA frozen and adding only lightweight trainable projection layers.

We follow a 3D full-resolution training plan at 1.5 mm isotropic spacing with patch size of $96 \times 128 \times 192$, batch size 2, and per-volume z-score normalization. Optimization uses SGD with an initial learning rate of 1×10^{-3} , weight decay 3×10^{-5} , and a polynomial learning-rate schedule over 1000 epochs. We report Dice and HD95 on the LiQA validation set.

4 Experiment

4.1 Experimental Settings and Datasets

We evaluate our approach on the LiQA GED4 validation split and the blinded test set. All experiments are implemented in PyTorch and run on a DGX cluster using a single NVIDIA A100 (80 GB) GPU per job.

LiQA Dataset We use the LiQA liver-fibrosis cohort comprising 610 patients collected across multiple centers and three MRI platforms—Philips Ingenia 3.0 T, Siemens Skyra 3.0 T, and Siemens Aera 1.5 T [6, 18, 29]. Each case includes T2-weighted imaging, diffusion-weighted imaging, and Gadolinium ethoxybenzyl diethylenetriamine pentaacetic acid (Gd-EOB-DTPA)-enhanced dynamic MRI covering the non-contrast, arterial, portal, delayed, and hepatobiliary phases. Data are provided in the Neuroimaging Informatics Technology Initiative (NIfTI) file format; phases other than the hepatobiliary phase may be absent, and no inter-phase or inter-subject pre-alignment is applied. Table 1 summarizes the splits across vendors and centers.

In this study, we train, validate, and test exclusively on hepatobiliary-phase (GED4) images. The training set contains 30 annotated cases in total (10 per center). The validation set includes 60 cases (20 per center), all with liver annotations. The held-out test split comprises 190 GED4 cases drawn from four vendor/center pairs; ground-truth liver annotations are withheld and used only for

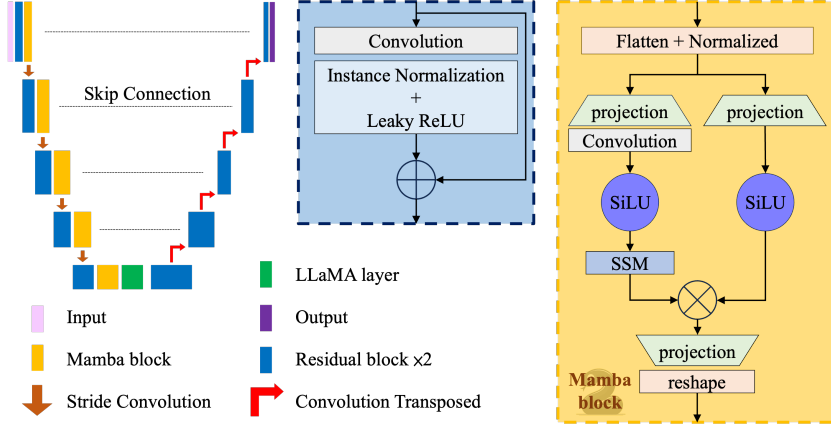


Fig. 1. Overview of LLaMba-Seg architecture: (a) Overall: The encoder stacks residual conv blocks with Mamba to couple local detail and linear-time global context. A frozen LLaMA-2 adapter at the bottleneck projects deepest features to LLaMA space, processes them, and projects back—replacing the deepest skip. The decoder upsamples with residual blocks, fusing this LLaMA-conditioned bottleneck with shallower skips; freezing LLaMA keeps parameters low while injecting strong priors. (b) Mamba block: The 3D feature map is flattened and layer-normalized, then split into a sequence path and a gating path. The two branches are combined via a Hadamard product, projected back to the feature width, reshaped to 3D, and added with a residual connection.

blinded evaluation. Notably, Vendor C is unseen during training/validation, providing an opportunity to assess cross-vendor generalization. This multi-vendor, multi-center design, combined with the lack of pre-registration, introduces substantial appearance and geometric variability, offering a realistic and challenging testbed for robust segmentation.

ATLAS Dataset The ATLAS (A Tumour and Liver Automatic Segmentation) dataset [23] was introduced in the MICCAI 2023 ATLAS Challenge to benchmark automated liver and tumor delineation on contrast-enhanced MRI and is used in this study as an external training dataset. It comprises 90 hepatocellular carcinoma (HCC) cases with liver and tumor annotations, released as a public CE-MRI resource with a 60/30 train-test split. Metadata (sequence/phase, resolution, scanner vendor) allow studies of domain variability and robustness. As one of the first openly available, fully annotated CE-MRI datasets for HCC segmentation, ATLAS supports evaluation under realistic clinical heterogeneity. In this study, we use only the training subset as external training data; no test data or labels are used.

AMOS Dataset We incorporate external supervision from the MICCAI 2022 AMOS (Multi-Modality Abdominal Multi-Organ Segmentation) dataset [13],

Table 1. LiQA dataset [6, 18, 29] composition across training, validation, and test splits. Test annotations are withheld by the challenge organizers for blinded evaluation.

Split	Vendor	Center	#Cases	#Annotations
Training	A	A	130	10
	B	B1	170	10
	B	B2	60	10
Validation	A	A	20	20
	B	B1	20	20
	B	B2	20	20
Testing	A	A	40	–
	B	B1	40	–
	B	B2	40	–
	C	C	70	–

Table 2. Network comparison on LiQA GED4 validation dataset (higher Dice is better; lower HD is better). Best in bold.

Network	Dice $\uparrow$	HD (mm) $\downarrow$
U-MambaBot	0.9603	26.53
U-MambaEnc	0.9684	22.63
U-MambaBot + LLaMA	0.9686	22.03
LLaMba-Seg (U-MambaEnc + LLaMA)	0.9700	21.34

Table 3. LiQA GED4 blind-test results using our LLaMba-Seg model. Higher Dice is better; lower HD is better.

Split	Dice $\uparrow$	HD (mm) $\downarrow$
in-distribution cases (ID)	0.9630	25.75
out-of-distribution cases (OOD)	0.9716	16.41

which contains 600 3D abdominal scans (500 CT, 100 MRI) with voxelwise annotations for 15 organs. Following the official Task 2 partitioning (300 CT + 60 MRI for training), we use the 60 MRI training cases as external training data to match our MRI target domain; all CT scans and the AMOS validation/test splits are excluded. Volumes are provided in NIfTI format, and from the multi-organ labels, we retain only the liver mask for supervision.

4.2 Evaluation metrics

We evaluate liver segmentation with two standard metrics: the Dice coefficient and the Hausdorff Distance (HD). Dice measures volumetric overlap between the prediction $\hat{Y}$ and reference Y , $\text{Dice} = \frac{2|\hat{Y} \cap Y|}{|\hat{Y}| + |Y|}$, with higher values indicating better agreement. HD (reported in millimeters) quantifies the worst-case boundary discrepancy as the maximum surface-to-surface distance between the predicted and reference masks; lower values indicate more precise boundary alignment.

Table 4. Per-case liver segmentation performance on five LiQA GED4 validation cases (the same cases shown in Fig. 2). Metrics are Dice ($\uparrow$) and symmetric Hausdorff distance in millimetres, HD ($\downarrow$).

	Case 1	Case 2	Case 3	Case 4	Case 5
Dice $\uparrow$	0.98285	0.98305	0.98067	0.97959	0.98020
HD (mm) $\downarrow$	22.299	24.047	41.024	162.651	17.685

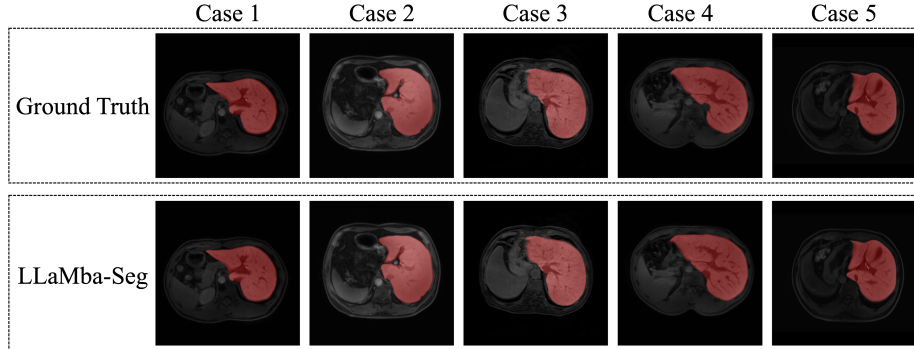


Fig. 2. Qualitative liver segmentation on five LiQA GED4 cases. Columns correspond to Cases 1–5 in Table 4. Top row: ground-truth masks; bottom row: LLaMba-Seg predictions. Red overlays denote the liver on the hepatobiliary-phase MR slices.

4.3 Results

Table 2 compares four configurations on LiQA dataset: U-Mamba with Mamba blocks in all encoder (U-MambaEnc) or only at the bottleneck (U-MambaBot), each with and without a frozen LLaMA-2 (7B) bottleneck adapter. Adding LLaMA improves both variants: for U-MambaEnc, Dice increases from 0.9684 to 0.9700 and HD drops from 22.63 to 21.34 mm. For U-MambaBot, Dice rises from 0.9603 to 0.9686 and HD decreases from 26.53 to 22.03 mm. While the LLaMA adapter yields larger relative gains on the bottleneck-only design, LLaMba-Seg remains the top performer overall on both metrics.

We also conduct a comprehensive evaluation of the challenge testing set, explicitly separating in-distribution (ID) cases—originating from vendors and imaging centers included during training (Vendors A, B1, and B2)—from out-of-distribution (OOD) cases collected from a completely held-out site (Vendor C). This setup enables us to rigorously assess the model’s robustness to variations in imaging protocols, scanner hardware, and acquisition environments. As summarized in Table 3, the model maintains strong overall performance on the ID subset while demonstrating even better results on the OOD subset, highlighting its ability to generalize across clinical sites.

Specifically, the OOD cases exhibit a higher Dice score (+0.0086) compared to ID cases, indicating more accurate volumetric overlap between predicted and ground-truth segmentation, despite differences in acquisition settings. Furthermore, the OOD set achieves a substantially lower HD by -9.34 mm, reflecting

more precise boundary delineation and fewer extreme segmentation errors on unseen sites. This is particularly noteworthy given that OOD data typically present additional challenges, such as shifts in image intensity distributions, heterogeneous contrast properties, and center-specific acquisition artifacts.

Figure 2 presents overlays for five cases (Cases 1–5 in Table 4). In each column, the top row displays the expert-annotated ground truth segmentations, while the bottom row shows the predicted masks generated from the proposed LLaMba-Seg model. Regions marked in red denote the segmented liver boundaries. Overall, the predicted contours visually align well with the true hepatic boundaries, demonstrating the model’s ability to adapt to diverse liver morphologies, varying contrast levels, and challenging anatomical contexts. Importantly, while the model achieves consistently high overlap across cases, subtle differences highlight the complementary insights provided by different evaluation metrics. For instance, Case 4 attains a high Dice (0.9796) yet a large HD (162.65 mm) due to a tiny distant false positive. This illustrates HD’s heightened sensitivity to even small, isolated false-positive regions, which may be clinically less significant but still strongly influence the boundary-based metric. Such observations emphasize the importance of jointly evaluating volumetric similarity and boundary accuracy for a comprehensive assessment of segmentation quality.

5 Conclusion

We introduced LLaMba-Seg, a U-Mamba encoder–decoder with a frozen LLaMA adapter at the bottleneck. By projecting the encoder features into LLaMA’s hidden space and back, the model injects contextual priors with minimal additional trainable parameters, complementing U-Net–style skips and improving boundary fidelity. On the LiQA hepatobiliary-phase cohort, our best configuration achieved a Dice of 0.9700 and HD95 of 21.34 mm, surpassing the original U-Mamba and both bottleneck variants, with robust generalization across multiple vendors/centers, including an unseen site. **Limitations of our method** include the use of a frozen, non-medically pretrained LLM, sequence-length overhead from bottleneck tokenization, and reliance on a single phase at 1.5 mm isotropic resolution. **Future work** will explore lightweight adapter tuning with clinical corpora, self-supervised pretraining for the Mamba encoder, multi-phase fusion, domain and test-time adaptation, uncertainty estimation for quality control, and efficiency improvements via tokenization or distillation to remove the LLaMA adapter at inference.

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